

George Ma

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Computer Science; GPA: 3.93

Expected June 2027

Relevant Courses: Design & Analysis of Algorithms, Advanced Data Structures, Data Structures & Algorithms, Systems Programming, Database Systems, Object-Oriented Programming, Computer Organization, Data Science Theory

TECHNICAL SKILLS

Languages: Go, Python, Java, C/C++, SQL, TypeScript, JavaScript, Bash/Shell, Swift, HTML, CSS, ARM Assembly

Libraries/Frameworks: FastAPI, Flask, Node.js, Express.js, PyTorch, scikit-learn, LiteLLM, React, React Native

Cloud/Infra: AWS, GCP, Kubernetes, Docker/Podman, Terraform, Jenkins, ArgoCD, Kafka, PostgreSQL/MySQL, Git

EXPERIENCE

Capital One

McLean, VA

Incoming Software Engineer Intern – DevOps/SRE

June 2026 – August 2026

IBM

San Jose, CA

Software Engineer Intern – Automation and AI (Project Bob)

February 2026 – May 2026

- Built a React Native/Expo mobile AI assistant with Flatlist chat rendering optimization for **7k+ internal users**
- Engineered **OAuth/OIDC** gateways in **Go** via PKCE, JWT/Redis-based sessions, and **OPA-based RBAC**
- Architected a **LiteLLM proxy** using FastAPI for an ACP-based Typescript agent framework, reducing workflow latency by **25%** via semantic routing for dynamic model selection across AWS Bedrock, WatsonX, and Mistral
- Automated deployment of backend, agent, and auth microservices to **OpenShift (K8s)** via Terraform/Jenkins
- Optimized Strimzi-managed **Kafka** pipelines using Prometheus/Grafana, reducing processing latency by **32%**

Praxie AI

San Francisco, CA

Software Engineer Intern – Mobile and Backend Infrastructure

April 2025 – February 2026

- Drove migration from Firebase to a containerized Express.js **REST API** on **Google Cloud Run**, provisioning infrastructure with **Terraform** and securing GitHub Actions CI/CD pipelines via Workload Identity Federation
- Architected scalable tournament and multi-player management in React Native using **PostGIS** geo-indexing and TanStack Query for optimized client-side caching, achieving **sub-1s search latency** for 300+ active users
- Built a fault-tolerant **NoSQL to Cloud SQL** (PostgreSQL) data migration and schema refactor pipeline using node-pg-migrate, denormalizing complex relational structures and improving data-fetching performance by **20%**

Alpha Kappa Psi @ UC San Diego

La Jolla, CA

Lead Full-Stack Engineer

December 2024 – January 2026

- Spearheaded a full-stack website migration from Wix to a production-grade **Next.js** and Supabase architecture, implementing **SSR** and optimized data access patterns to achieve **60% faster** page load times for **2,000+ MAU**
- Led Git-based Agile sprints with detailed documentation for scalable feature delivery and long-term maintenance
- Designed a normalized **PostgreSQL** schema serving dynamic content with **sub-1s latency** under concurrent load

Solana Center for Environmental Innovation

Encinitas, CA

Data Engineer Consultant

March 2025 – June 2025

- Developed an automated **Python ETL** pipeline (Pandas/NumPy), cutting 15+ hours of manual KPI reporting
- Automated model benchmarking (SARIMA, Prophet, XGBoost) for operational forecasts with **88.2% accuracy**

PROJECTS

Watchdog | [Demo](#)

August 2025 – September 2025

- Built an **AI-powered CI/CD** automation for Github PR reviews, linting, and security scans for 6+ languages
- Engineered 2 containerized **MCP microservices** using FastAPI and Express.js on **AWS ECS** and **Fargate**
- Implemented a parallelized smart-chunking algorithm for accurate, in-depth LLM reviews of complex code diffs
- Orchestrated multi-language, conditionally executed CI across 13+ toolchains, reducing build time by **80%**
- Enhanced user security and reliability by implementing graceful degradation and least-privilege token scoping

Pokemon Generator | [Demo](#) | [Website](#)

April 2025 – July 2025

- Created a full-stack, containerized web app generating custom Pokemon using Flask, Tailwind CSS, and PyTorch
- Architected a custom **PyTorch Conditional GAN** utilizing **CUDA** acceleration to generate unique 256x256 pixel Pokemon images from user inputs mapped to latent condition vectors via a **RESTful inference API**
- Implemented a relational PostgreSQL schema with SQLAlchemy ORM to map and store user's creations
- Developed a prediction service using **scikit-learn** RandomForestRegressors to simulate balanced game attributes